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Studierichting: Informatica
Oefeningenreeks 3A 8.25

$$f(x) = \frac{2x^2 - 4x + 1}{\sqrt{x}}$$

$$f'(x) = \frac{\sqrt{x}(4x - 4) - (2x^2 - 4x + 1) \frac{1}{2\sqrt{x}}}{x}$$

$$= \frac{\frac{(4x - 4)}{2\sqrt{x}} - \frac{2x^2 - 4x + 1}{2\sqrt{x}}}{x}$$

$$= \frac{8x^2 - 8x - 2x^2 + 4x - 1}{2\sqrt{x}}$$

$$= \frac{\frac{6x^2 - 4x - 1}{2\sqrt{x}}}{x}$$

$$= \frac{6x^2 - 4x - 1}{\sqrt{x^3}}$$

References